B. K. Birla College of Arts, Science and Commerce, Kalyan (W)

TYBSc Physics, Semester VI Special Theory of Relativity

Assignment 5

Unit 3 (Part A): The Geometric Representation of Space-Time

Name: _____ Roll No.: _____ Date: _____

Total marks: 60 **Each question carries 5 marks**

Instructions: Attempt all questions. Section A: theory (write to the point, with neat diagrams where useful). Section B: numerical problems and derivations (state formulae used, show all steps and give units). Sign convention for the interval: $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$; four-vectors $a^\mu = (a^0, \mathbf{a})$ with $a \cdot b = -a^0 b^0 + \mathbf{a} \cdot \mathbf{b}$.

Section A: Theory (6 × 5 = 30 marks)

Q1. What is a space-time diagram? Explain world lines and light lines, and show how the ct' and x' axes of a moving frame are drawn. How does the diagram display the relativity of simultaneity? **[5]**

Q2. Using space-time diagrams, explain time dilation and length contraction, and explain why the statement "each observer finds the other's clocks running slow" involves no contradiction. **[5]**

Q3. Explain the light cone of an event. Classify the separation of two events as timelike, spacelike or lightlike, and discuss for each the time order of the events and the possibility of a causal connection. **[5]**

Q4. What is a four-vector? Explain contravariant and covariant components, the metric tensor, the rank of a tensor and the summation convention. Why do laws written as tensor equations hold in all inertial frames? **[5]**

Q5. State the twin paradox. Explain why the situation of the two twins is not symmetric and how the paradox is resolved using proper time along world lines. **[5]**

Q6. State the principle of equivalence. Describe Einstein's elevator thought experiments and explain the consequences of the principle: the bending of light, the gravitational red shift and the geometric interpretation of gravity in general relativity. **[5]**

Section B: Numericals and Derivations (6 × 5 = 30 marks)

Q7. Prove that the interval $s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2 - c^2\Delta t^2$ is invariant under the Lorentz transformation. Event B occurs 6 light-years from event A and 10 years after it (in frame S). Find s^2, classify the interval, find the proper time between the events and the speed of the frame in which they occur at the same place. **[5]**

Q8. Event A is at the origin and event B is at $x = 900\,\text{m}$, $t = 2.0\,\mu\text{s}$. (a) Find s^2 and classify it. (b) Find the proper distance between the events and the speed of the frame in which they are simultaneous. (c) Find the time interval between them in a frame moving at $0.8c$ along x, and comment on the order. (d) Can A be the cause of B? **[5]**

Q9. Two four-vectors have components $A^\mu = (5, 4, 0, 0)$ and $B^\mu = (3, 1, 2, 2)$. (a) Write A_μ and B_μ. (b) Find $A \cdot A$, $B \cdot B$ and $A \cdot B$ and classify A and B. (c) Find the components of A^μ in a frame moving at $0.8c$ along x and verify that $A \cdot A$ is unchanged. **[5]**

Q10. (a) Show that the four-velocity $u^\mu = \gamma(c, \mathbf{v})$ satisfies $u_\mu u^\mu = -c^2$; check it for a particle moving at $0.6c$ along x. (b) Show that if a tensor is antisymmetric ($T^{\mu\nu} = -T^{\nu\mu}$) in one inertial frame, it is antisymmetric in every inertial frame. **[5]**

Q11. A twin travels to a star 6.0 light-years away (Earth frame) at $0.6c$ and returns at the same speed. (a) Find the time elapsed on the Earth and for the travelling twin, using the interval along each leg. (b) Each twin sends a light signal once a year (by their own clock). Using the Doppler factor, show that the number of signals each receives agrees with (a). **[5]**

Q12. Derive the gravitational red shift $\Delta\nu/\nu = -gh/c^2$ using the principle of equivalence. Calculate (a) the fractional shift in the Pound–Rebka experiment ($h = 22.5\,\mathrm{m}$), and (b) the red shift $z \approx GM/Rc^2$ and the shift of the Hα line (656.3 nm) from a white dwarf of mass $0.6M_\odot$ and radius 9000 km. **[5]**

Useful constants: $c = 3.0 \times 10^8$ m/s; $e = 1.6 \times 10^{-19}$ C; $m_e c^2 = 0.511$ MeV; $m_p c^2 = 938.3$ MeV; $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m; $1/(4\pi\varepsilon_0) = 9.0 \times 10^9$ N m^2/C^2; $G = 6.67 \times 10^{-11}$ N m^2/kg^2; $M_\odot = 1.99 \times 10^{30}$ kg; 1 ly $= 9.46 \times 10^{15}$ m; 1 pc $= 3.086 \times 10^{16}$ m $= 3.26$ ly; 1 yr $= 3.156 \times 10^7$ s; $b_{\mathrm{Wien}} = 2.898 \times 10^{-3}$ m K; $k_B = 8.62 \times 10^{-5}$ eV/K; $a = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}.